

Background

- Depression affects ~18.5% of U.S. adults (Lee et al., 2023)
- Physical limitations (e.g., difficulty walking) may increase depression risk
- Anxiety frequently co-occurs with depression and may worsen outcomes
- It is unclear whether anxiety frequency modifies the walking difficulty-depression association
- Gap: This study examines if the association varies by how often individuals feel worried, nervous, or anxious

Study Objectives

Research Question:

- Among U.S. adults aged 20 years and older, is greater difficulty walking or climbing steps associated with higher depression severity, and does this association vary by how often individuals feel worried, nervous, or anxious?

Hypothesis:

- Greater walking difficulty is associated with higher depression severity (higher PHQ-9 scores and greater likelihood of probable depression)
- This association is stronger among those who feel anxious daily or weekly (frequent) compared to those who feel anxious less often (infrequent: monthly, a few times a year, never)

Methods

Study design and sample. This cross-sectional study used data from NHANES 2021-2023. The analytic sample included 7,809 U.S. adults aged 20 years and older.

Variables and analysis. Depression severity was measured using the PHQ-9 (continuous 0-27; binary ≥ 10 for probable depression). The exposure was self-reported difficulty walking or climbing steps. Univariable analyses described the sample. Bivariable associations were assessed using t-tests, chi-square tests, and ANOVA.

Results

Univariable analysis (Sample Description). The sample included 7,809 U.S. adults aged 20 years and older. Mean age was 53.6 years, and 55.3% were female. Most were non-Hispanic White (58.3%) and had at least some college education (64.2%). One-third (33.7%) reported any walking difficulty. The mean PHQ-9 score was 4.0 (SD = 4.7), and 12.7% met depression criteria (PHQ-9 ≥ 10).

Bivariable analysis. Walking difficulty was significantly associated with higher mean PHQ-9 scores (5.5 vs. 3.3, $p < 0.001$) and a higher prevalence of depression (52.8% vs. 47.2%, $p < 0.001$). Diabetes met confounder criteria ($p < 0.001$), but high cholesterol did not ($p = 0.277$). Depression prevalence was 25.0% with frequent anxiety vs. 4.1% with infrequent anxiety ($p < 0.001$), suggesting that anxiety may modify the association.

TABLE 1: Univariable Analysis of U.S. Adults Aged 20 Years and Older, National Health and Nutrition Examination Survey (NHANES), United States, 2021-2023 (n = 7809)

Sociodemographic characteristics	Total N (%)
Gender (N = 7809)	
Male	3488 (44.7%)
Female	4321 (55.3%)
Age (years) (N = 7809)	
Mean (SD, range)	53.6 (17.5, 20 - 80)
Median age (IQR)	57 (38 - 68)
Education level (N = 7783)	
Less than High School	1039 (13.3%)
High School Graduate/GED	1749 (22.5%)
Some College or College Graduate	4995 (64.2%)
Ratio of family income to poverty (N = 6489)	
≤ 1	1044 (16.1%)
1	5445 (83.9%)
Depression (PHQ-9) Score (continuous) (N = 5210)	
Mean (SD, range)	4.0 (4.7, 0 - 26)
Median age (IQR)	2 (1 - 6)
Depression (N = 5210)	
No	4549 (87.3%)
Yes	661 (12.7%)
Difficulty Walking or Climbing Steps (N = 7797)	
No Difficulty	5167 (66.3%)
Any Difficulty	2630 (33.7%)
High Cholesterol (N = 7755)	
Yes	3052 (39.4%)
No	4703 (60.6%)
Diabetes Condition (N = 7806)	
Yes	1070 (13.7%)
No	6736 (86.3%)
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)	
Infrequent	4656 (59.8%)
Frequent	3129 (40.2%)

SD = standard deviation; IQR = interquartile range

TABLE 2: Bivariable associations between walking difficulty and depression (n = 7,809)

Sociodemographic characteristics	No N (%)	Yes N (%)	p-value
Difficulty Walking or Climbing Steps (N = 7797)			
No Difficulty	3175 (91.1%)	312 (8.9%)	<0.001 ²
Any Difficulty	1372 (79.7%)	349 (20.3%)	
High Cholesterol (N = 7755)			
Yes	1934 (87.9%)	265 (12.1%)	<0.277 ²
No	2595 (86.9%)	390 (13.1%)	
Diabetes Condition (N = 7806)			
Yes	579 (80.9%)	137 (19.1%)	<0.001 ²
No	3970 (88.3%)	524 (11.7%)	
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)			
Infrequent	2945 (95.9%)	127 (4.1%)	<0.001 ²
Frequent	1599 (75.0%)	534 (25.0%)	

¹p-value for simple logistic regression; ²p-value for chi-square test or Fishers exact test; ³p-value for t-test; ⁴p-value for ANOVA

Discussion

- Main finding:** Walking difficulty is associated with higher depression severity; anxiety frequency may modify this association.
- Strengths:** Large, nationally representative sample (NHANES); validated PHQ-9 measure.
- Limitations:** Cross-sectional design (cannot establish causality); 33% missing PHQ-9 data; self-reported measures may introduce recall bias.
- Future research:** Longitudinal studies needed to examine temporality and whether improving walking ability reduces depression.